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Jet stream and long waves in a steady rotating-dishpan experiment: structure of the circulation

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Summary

The three-dimensional structure of the flow of water in a steady rotating three-wave dishpan experiment is computed and compared with the atmosphere.

At first the experiment and methods of comparison between dishpan and atmosphere are described. Then motion and vorticity at the top surface are shown; trajectories leading from equatorial to polar rim of the pan are found. Next the three-dimensional temperature field is determined, and from this the geostrophic motion at all depths.

The flow pattern resembles the basic features of the middle-latitude atmosphere in a remarkable way. Mobile long waves in a jet stream near the top overlie cyclonic and anticyclonic vortices farther down. There is, however, neither the counterpart of tropopause nor stratosphere in the temperature field, indicating that these features are not necessary for development of the basic flow pattern. Further, the distribution of momentum sources and sinks differs from that of the atmosphere, as momentum is given off to the inner (and possibly to the outer) cylinder, while most of the bottom acts as source. This throws doubt on the importance of momentum balance considerations for explanation of the principal flow features.

Vertical motions are calculated in the interior of the fluid with the adiabatic assumption. Values again compare well with those of the atmosphere as does the distribution of centres of ascent and descent relative to long waves and cyclones. A positive correlation between temperature and vertical motion exists along the jet axis, hence there is release of potential energy. If averaging is performed along the jet-stream axis, one finds that the centre of ascent lies equatorward, and the centre of descent poleward of the axis. If *zonal* averaging is performed, however, the centre of ascent appears north of the zonal wind maximum, and the centre of descent on its equatorward side. This calculation uncovers a weakness of zonal averaging as a tool in studying the general circulation.

Absolute vorticity is calculated on constant temperature surfaces in the interior of the liquid. It is found that the theorem of conservation of absolute vorticity is nearly fulfilled in the jet-stream layer with wave motion; in the lower layer with vortices, validity of the theorem of conservation of potential vorticity is indicated qualitatively. The regions of convergence and divergence, as inferred from the vorticity changes along the relative stream-lines, agree qualitatively with the vertical motion distribution.

1. Introduction

The remarkable similarities noted between the circulations of the upper troposphere and the surface of the water in rotating-dishpan experiments (Fultz 1950, 1952, 1956, Corn and Fultz 1955) have encouraged extension of these experiments to determine the three-dimensional structure of the dishpan circulations. At the present stage of technical development in these experiments this is made possible through the discovery by Hide (1953, 1956) and Fultz (1952, 1956) that when the liquid is contained between two con-concentric cylinders, the inner of which is a cold source and the outer a heat source, steady-state wave motion may be developed over a range of experimental conditions. In a coordinate system moving at the rate of propagation of the waves the field distribution of motion, temperature, etc., remains unchanged. Because of this steady state one can measure the entire three-dimensional temperature field with only one or two measuring elements which are placed at a number of different locations. 'Thermograph' traces of the time variation can then be converted to longitude variation.

For a first try a case with three steady long waves was chosen. As shown by Hide (1956) and Fultz *et al.* (1956), the surface-flow pattern for any given pan size, inner cold-source cylinder size, and depth, is determined by the rate of rotation of the pan and the rate of energy supply. In the experiment to be described the pan had a radius of 15·6 cm, the water depth was 4 cm and the rate of revolution 0·30 sec^{-1}. A heat source with nominal intensity of 150 w was applied at the equatorial rim, and a central cylindrical cold source with radius of 6·5 cm provided for heat balance. Actual heat transport to the cold source was measured at about 64 w; this is probably about 20 per cent too high because of extraneous gains at the cold source. The mean temperature of the water in the pan was about 19·5°C and the temperature of the circulating water in the cold source about 10°C. Given these physical conditions a pattern of three long waves on a continuous jet stream emerges when steady state is attained. Fig. 1 (Plate IV) shows the top-surface flow pattern, made visible by photography of aluminium powder floating at the surface.

The objectives of the investigation were (a) to determine the structure of the circulation in the dishpan and to compare it with the atmosphere; (b) to calculate the heat and momentum budgets; (c) to ascertain, as far as possible, the mechanism governing maintenance of long waves and jet stream, and their role in the general circulation.

This report deals mainly with the first of these objectives.

2. Data

The observations consisted of (a) surface-velocity measurements made from the original of Fig. 1 and (b) temperature measurements made at different radii and depths so that the three-dimensional temperature field could be analysed. Fultz *et al.* (1956) have discussed the details in making these measurements.

The three waves of Fig. 1 are quite symmetrical; thus only one wave need be analysed. Some subordinate microstructure was present, evident especially along the equatorial margin of the pan. This microstructure, an unsteady part of the flow, was eliminated by averaging velocity and temperature fields over all three waves.

Some of the features of the flow encountered during the analysis were subsequently checked and verified at least qualitatively by repeating the experiment.

3. Comparison of dishpan and atmosphere

If circulations on bodies of such different size as dishpan and earth are to be compared successfully, the variables must be properly selected. Rossby (1947) accomplished this for the motion variables when he found from data on earth and sun that the equatorial velocity of a rotating body provides a pertinent reference velocity, in terms of which other velocities may be expressed. One may think of relative velocities, for example, expressed non-dimensionally as ratios to this reference speed or, what is equivalent, expressed with the equatorial velocity as the unit.

Different types of experiment have amply verified the validity of Rossby's deduction. Given the relative wind u, the angular rotation ω and the radius r_0 of any rotating system, the ratio $R = u/r_0\,\omega$ expresses the wind speed as a fraction of the equatorial velocity $C_e = r_0\,\omega$. This method of representing the wind is one of a number of more or less equivalent choices of similarity variables; it is particularly convenient for practical reasons. Fultz (1951) and others have called R the 'Rossby number' when u is a representative relative velocity. If we express $R = \dfrac{u}{r_0\,\omega} = \dfrac{u^2}{r_0\,u\omega} = \dfrac{u^2}{r_0}\Big/u\omega$, we see that R is also a measure of the 'convective' acceleration as a fraction of the Coriolis acceleration. This is the really significant aspect for dynamic similarity of the experiments to the atmosphere since it expresses one of the most important typical force ratios involved, i.e., of inertia forces to Coriolis forces. The interpretation of u^2/r_0 holds as long as r_0 is of the order of a typical wave length or other typical horizontal scale for the motion (Charney 1948, Fultz and Frenzen 1955).

The smaller R, the more nearly geostrophic will be the flow. In the present case, the highest value of R is 0·2. Hence the motion to be treated is quasi-geostrophic, though 0·2 is attained over relative areas considerably larger than in the high troposphere. Consequently the experimental system is somewhat more ageostrophic than the atmosphere. We may note here that the reference velocity $r_0\,\omega$ for the earth is about 1000 mi hr^{-1} and also that the experimental work mentioned earlier, at least for some broad aspects of the experiments, shows that the geometrical distortion from spherical shape does not vitiate direct comparison of velocity fields, etc., on the kinematic basis.

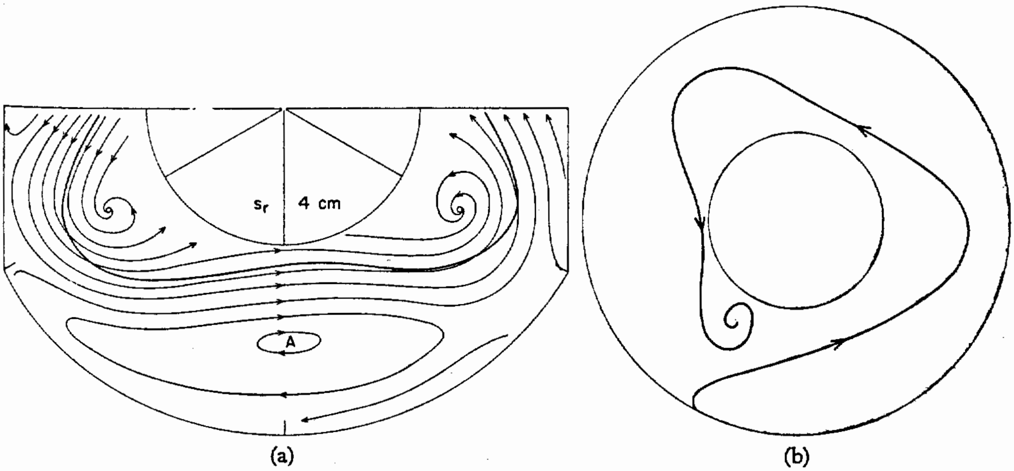
While it is not the intention of the present discussion to consider in detail the question of model similarity for the experiment, it is well to note also the similarity situation with respect to at least two of the parameters that depend on the temperature and density fields driving the whole system. The driving force, due to the average horizontal temperature gradient, is represented in a significant way by the resulting thermal wind difference, u_T, between top and bottom. If this wind is expressed in terms of $r_0\,\omega$ as $u'_T = \dfrac{u_T}{r_0\,\omega}$, we obtain the 'thermal Rossby number' of the system (Fultz 1956, Fultz *et al.* 1956). Its value for this experiment is about 0·18 and consequently is substantially higher than typical atmospheric values of middle latitudes which lie near 0·03. This is consistent with the higher 'kinematic Rossby number' in the jet (0·2) and the somewhat less geostrophic character of the motion previously mentioned.

The second parameter of particular interest is the large-scale Richardson number that appears, for example, in the baroclinic wave theories of Eady (1949) and others. This number is a ratio of vertical stability to the square of the vertical wind shear (for which we use the thermal wind shear). In the experiment its value is about 8, compared to a typical range of from 40 to 140 in middle latitudes.

The detailed implications of these and differences in other parameters are in many cases not yet clear. For general discussions of the methods of analysis see, e.g., Langhaar (1951) and for the meteorological experiments, Fultz *et al.* (1956).

4. Top surface flow

In general, the observed motions will be described in a coordinate system rotating with the speed of the pan. However, much importance attaches to the motion of the water *relative* to long waves and jet stream; in particular there is the question whether fluid



(a) (b)

Figure 2. (a) Relative streamlines on top surface. Heavy solid line is jet-stream axis. (b) Typical path followed by water from heat source to cold source on top surface.

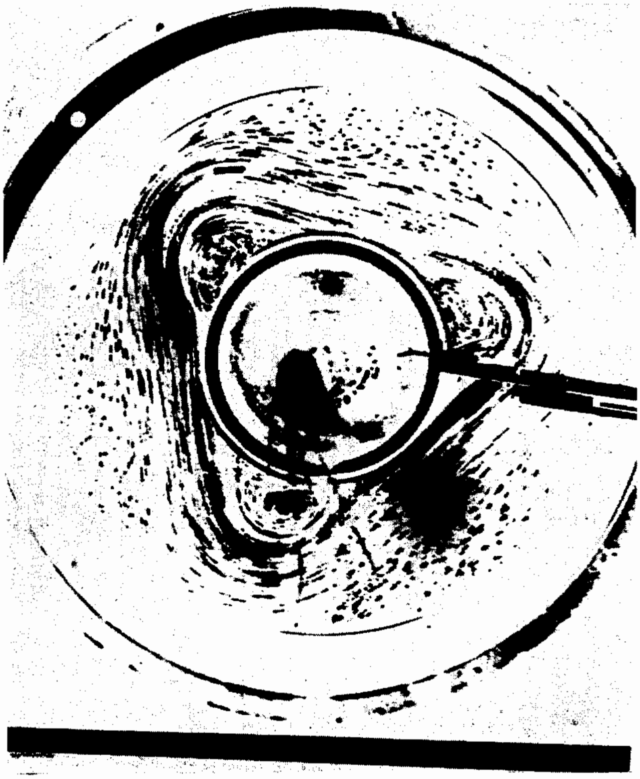


Figure 1. Photograph of top-surface flow.

passes across the jet-stream axis to produce a so-called 'cross-stream circulation.' The waves seen in Fig. 1 travel at a regular rate of 25° longitude per revolution (or per day) relative to the pan. They thus complete a circuit around the pan in fifteen revolutions, travelling with more than twice the speed of 'short' waves in the upper westerlies of the troposphere. Because of the regular progression, the linear wave speed may be subtracted vectorially from all winds, allowing for the radial variation of this speed. The residual is the motion *relative* to the waves.

Fig. 2 (a) shows relative streamlines at the surface computed in this manner. We observe closed paths in the ridge, a sink near the cold source in the trough and a source at the equatorial rim, also in the trough. From this source the relative streamlines spiral inward gradually. A typical streamline reaches the polar sink after passing through all three waves, i.e., it terminates in the same trough in which it starts (Fig. 2 (b)). Thus water does cross the jet stream axis from equatorial to polar side. All crossing takes place in the troughline and just to its east. This feature of the analysis – crossing of the jet – however is rather delicate and could easily be affected by experimental errors in the velocity measurements. Confirmation was obtained in repeat experiments subsequent to the analysis by using tracer-ink bands in the interior of the liquid. The motion of these bands is not subject to some of the difficulties of the aluminium powder measurements (Fultz *et al* 1956).

The field of wind speed depicts the jet stream clearly (Fig. 3). A broad band with speed of 20 per cent (comparable to 200 mi hr^{-1}) extends almost uniformly along the jet axis. Although two speed centres have been drawn, no importance can be attached to this feature which is well within the margin of error of the observations. As in the atmosphere, the cyclonic shear greatly exceeds the anti-cyclonic shear. Generally, however, the shears are much weaker than those observed on weather maps. This becomes evident when the field of absolute vorticity is computed (Fig. 4).

The absolute vorticity $\xi_a = 2\omega + \xi_r$, where ξ_r is the relative vorticity. In order to reduce ξ_a to values comparable to the atmosphere, we divide by the angular rotation of the pan. This is consistent with the method of expressing velocities in terms of $r_0\,\omega$. Denoting non-dimensional quantities with primes, $\xi_a' = 2 + \xi_r'$ where the number 2 denotes the value of the Coriolis parameter. In Fig. 4 the line, $\xi_a' = 2$, closely parallels the jet-stream axis, but its amplitude is greater because of the variation of streamline curvature along the axis. A band of minimum vorticity lies equatorward of the core, but values are only some 40 per cent less than the Coriolis parameter; zero absolute vorticity is approached nowhere. Similarly, the highest cyclonic vorticities on the poleward side of the axis exceed the Coriolis parameter by only 120 per cent. As demonstrated by

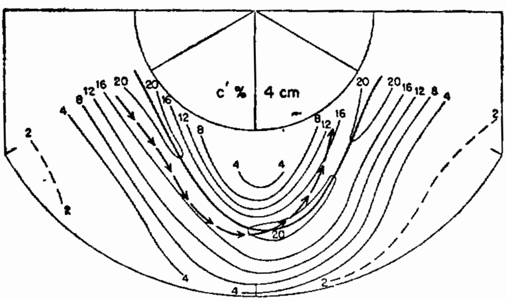


Figure 3. Speed of flow on top surface (per cent of equatorial speed of pan). Heavy solid line marks jet-stream axis; heavy dashed line with arrows shows part of a relative streamline crossing axis.

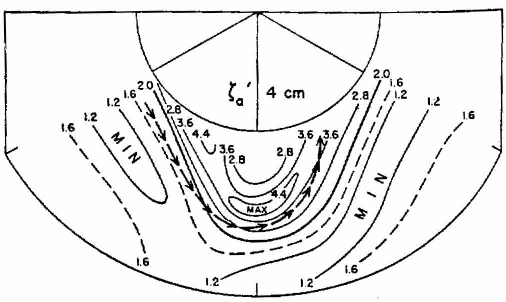


Figure 4. Absolute vorticity on top surface (non-dimensional). Vorticity is given by Coriolis parameter along heavy line with value 2·0. Heavy dashed line as in Fig. 3.

Fultz *et al.* (1956) vorticity fields that resemble those of the atmosphere more exactly are generated in experiments with higher rotation rates.

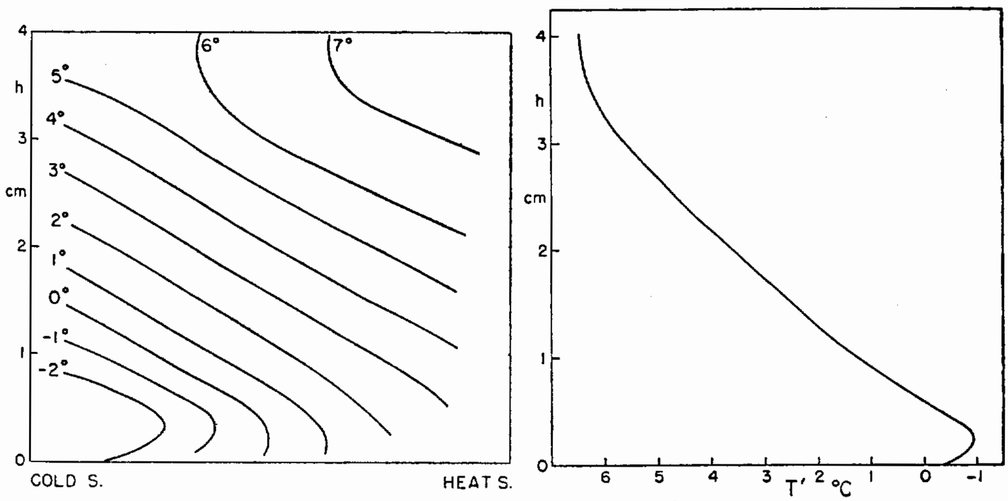
A portion of a relative streamline from Fig. 2 which crosses the jet has been entered in Figs. 3 and 4. As the water approaches the core of the jet it accelerates, then decelerates strongly after crossing the axis. The motion is directed toward increasing vorticity; as will be shown later, descent takes place below the area where the vorticity increase occurs.

5. Temperature field

A mean radial cross-section of temperature (temperatures are given as departures from 16°C), (Fig. 5a), obtained by averaging over all meridians, shows a poleward directed temperature gradient at all heights. This gradient is strongest at the bottom (6°C) and diminishes upward, with temperatures generally increasing with height, as is the case with potential temperature in the atmosphere. At the top we still observe a temperature drop of approximately half the strength of that at the bottom : tropopause and stratosphere do not exist. This suggests that these features of the atmospheric temperature field are not needed for production and maintenance of a jet stream and long waves.

In the interior of the water the stratification is very stable; even so, the stability is less than in the atmosphere as shown by the low Richardson number. Near top and bottom the vertical temperature gradient reverses, hence a convective regime prevails in relatively thin layers near these boundaries. At the bottom this is owing to the fact that the heat source at the rim was not insulated, so that heating also spread along the bottom. At the top some evaporation took place under the laboratory air conditions prevailing at the time of the measurements. The evaporation, measured from the volume loss, was sufficient to produce a heat loss rate of about 8 w.

Fig. 5 (a) may be averaged radially to yield a standard temperature-depth curve (Fig. 5 (b)). Between 0·5 and 3·5 cm the mean vertical temperature gradient is linear and amounts to 1°C per 0·5 cm.



(a) (b)

Figure 5. (a) Zonally-averaged distribution of temperature (departure from 16°C) with radius and height above bottom (cm). (b) Standard temperature-height curve for experiment.

Looking at a selection of horizontal temperature charts (Figs. 6 (a), (b), (c) and (d)) we observe in some measure what Rossby (1942) once termed a quasi-uniform swaying north and south of a very deep atmospheric layer. The phase shift with depth is slight, but the amplitude of the isotherm waves decreases downward. At the top surface the influence of evaporation is pronounced as seen from comparison with Fig. 2; in the region of closed anticyclonic streamlines in the ridge, where the same water circulates slowly for a long time, a secondary minimum of temperature is located. The inward-spiralling streamlines pass toward lower temperatures; therefore there is relative advection of warm water. Evaporation and convection in the unstable top layer provide for maintenance of the steady temperature field.

6. Pressure and wind-field

Because the density of the water is a function of temperature alone, under the conditions of the present experiment, i.e., the pressure compressibility of the water is very low, the vertical pressure distribution can be calculated in any column given the bottom pressure or configuration of the top surface. Neither of these data was available, although it is hoped to obtain at least one of them in future experiments. As the only remaining alternative, the surface configuration was computed from the wind-field. This was done from the complete equation of motion, and also geostrophically. The difference between the two contour fields was small; since later the contour fields at higher pressures or greater depth were used to calculate geostrophic flow, the geostrophic contour field was finally adopted also for the top surface, to be consistent. It should be noted that the contour field in question is that of heights relative to the level surfaces of 'apparent' gravity for the rotating-pan coordinate system. These surfaces are the same as those of the equilibrium paraboloid $z = z_0 + \dfrac{\omega^2 r^2}{2g}$ of a free surface of liquid rotating rigidly with the pan.

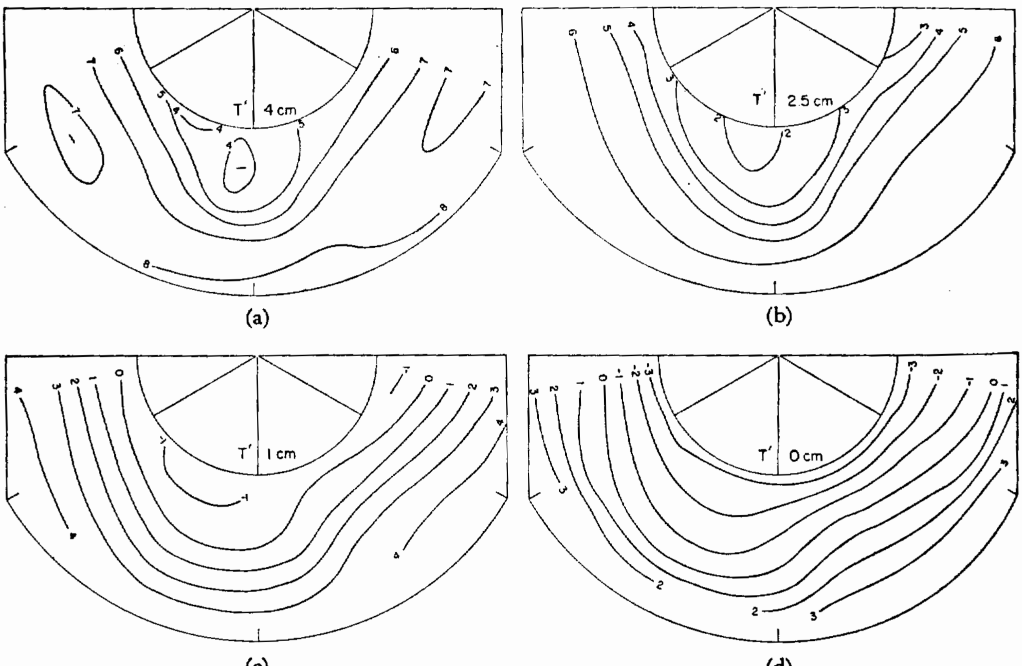


Figure 6. (a), (b), (c) and (d): Isotherms (departure from 16°C) at four selected levels; heights are indicated on diagrams.

For the low rotation value of 0·30 sec^{-1}, the rise at the rim of a level surface from a plane tangent to it at the axis is only 0·011 cm. Since the location of temperature measuring elements was known only to about 1 mm there was no need to allow for the curvature of the level surfaces in locating temperature points for calculations.

The geostrophic flow (c_g) on a constant-pressure surface is calculated from the well-known formula $f\,c_g = -g\dfrac{\partial D}{\partial n}$, where g is the acceleration of gravity, D the altimeter correction and n the horizontal axis normal to the wind, taken positive toward the left of the wind looking downstream. For non-dimensional calculation this equation becomes, after division by $r_0\,\omega^2$,

$$c_g' = -\tfrac{1}{2}\frac{\partial D'}{\partial n'}, \qquad (1)$$

where $n' = n/r_0$ and $D' = gD/C_e^2$. Generally, we shall express distances as fractions of r_0 such as n'. The quantity D' is obtained from the hydrostatic equation. Given the pressure interval δp between two isobaric surfaces, the specific volume α and the thickness between the isobaric surfaces δz, $\alpha\,\delta p = -g\,\delta z$. In standard water, denoted by the subscript 's,' $\alpha_s\,\delta p = -g\,\delta z_s$. Then $\alpha/\alpha_s = \delta z/\delta z_s$. Solving for δz and integrating from the top down, the height of any constant pressure surface

$$h = h_{\text{top}} - \bar{\alpha}/\alpha_s\,\delta z_s, \qquad (2)$$

where $\bar{\alpha}$ is the pressure-averaged specific volume. This equation assumes a very convenient form if we choose $\alpha_s = 1$, $\delta z_s = 1$. The specific volume $\bar{\alpha}$ is obtained from Dorsey's (1940) Tables on the expansion of water. Since $D = h - h_s$, the field of D' is readily determined from Eq. 2, given the topography of the top surface.

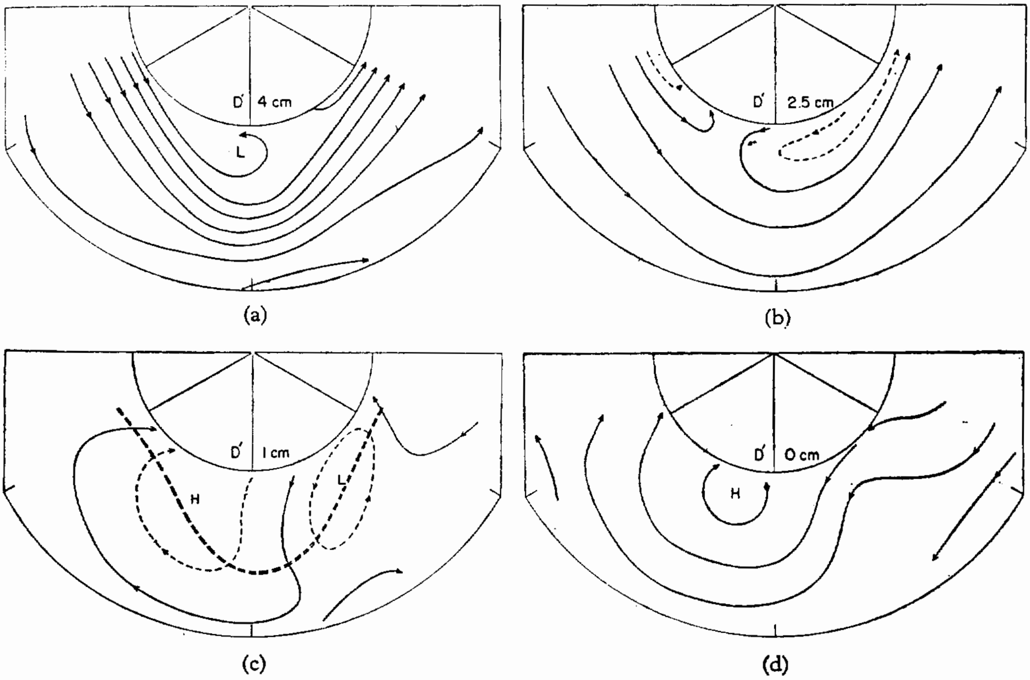


Figure 7. (a), (b), (c) and (d): Contours on four selected isobaric surfaces (D′) shown as stream function of geostrophic velocity. Double-dashed line at 1 cm is jet-stream axis at top surface (heights are indicated on diagrams).

Figs. 7 (a), (b), (c) and (d) contain the analysis of selected constant-pressure charts, expressed as a stream function of the geostrophic wind. (Since the altimeter corrections involved have the order of magnitude of microns, it is still permissible to refer to the charts in terms of constant height for convenience). In general, the lines could be drawn with ease but some smoothing proved necessary near 1 cm where gradients are weak and where the pattern changes rapidly with depth.

The jet stream decays downward and goes over into a system of vortices below 2 cm. At 1 cm the jet-stream axis has been entered for reference. We observe a cyclone and an anticyclone directly under the jet, with centres near southwest and northwest inflection points of the surface current. This then is the analogue of the atmospheric sea-level chart, but it is not the bottom chart in the dishpan where we find a sinusoidal easterly current, much as we would obtain in the atmosphere in middle latitudes after downward extrapolation to, say, 1200 mb.

Owing to the influence of the frictional boundary layer there were some winds at the bottom in the region under and just ahead of the upper trough not indicated by the geostrophic analysis. These were mainly north-westerly winds that occurred in a narrow sector behind a shallow cold front, which appeared to lie on the average 10° longitude ahead of the upper trough. This front extended only slightly above 0·5 cm height; the evidence for its existence was given by much sharper breaks downward in the temperature traces than occur anywhere else even where, in the upper layers, larger ranges of temperature were observed as the waves pass. Both this front and some smaller temperature breaks farther west which resemble secondary cold fronts, were somewhat variable in exact location and were smoothed out in the averaged analysis. The *local and shallow nature* of the near-discontinuity surface compared to the upper-temperature gradients in the jet certainly *suggests a minor role for the front in the overall mechanics of the waves*; perhaps it is no more than a minor result of the distribution of temperature advection.

The differences between dishpan and atmosphere arising from the predominant bottom easterlies will be discussed in detail in another report. Here it should merely be mentioned that they are explained readily from momentum-balance considerations. All interaction between atmosphere and earth must take place at the ground, while in this dishpan experiment the water loses momentum to the inner and possibly also the outer

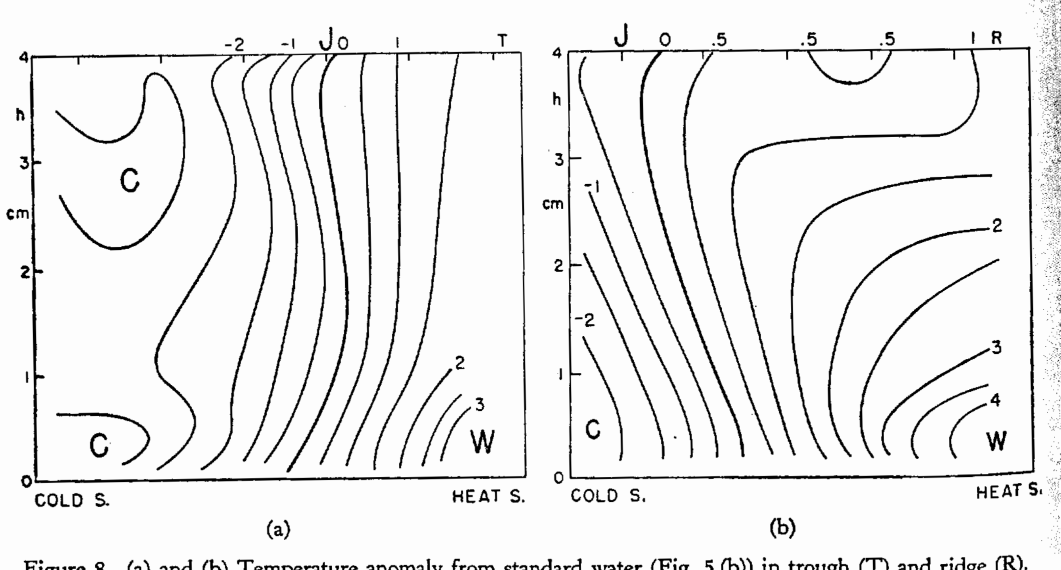
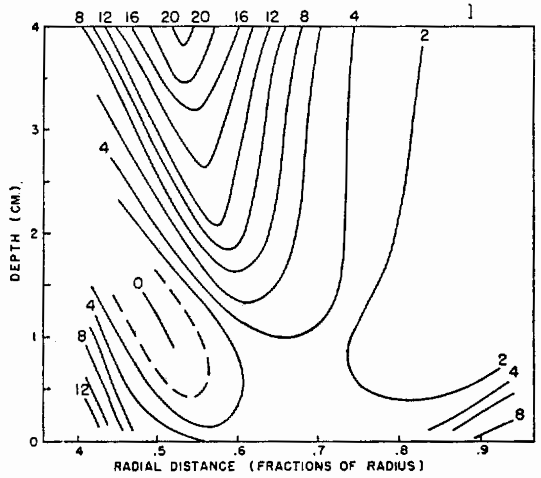


Figure 8. (a) and (b) Temperature anomaly from standard water (Fig. 5 (b)) in trough (T) and ridge (R).

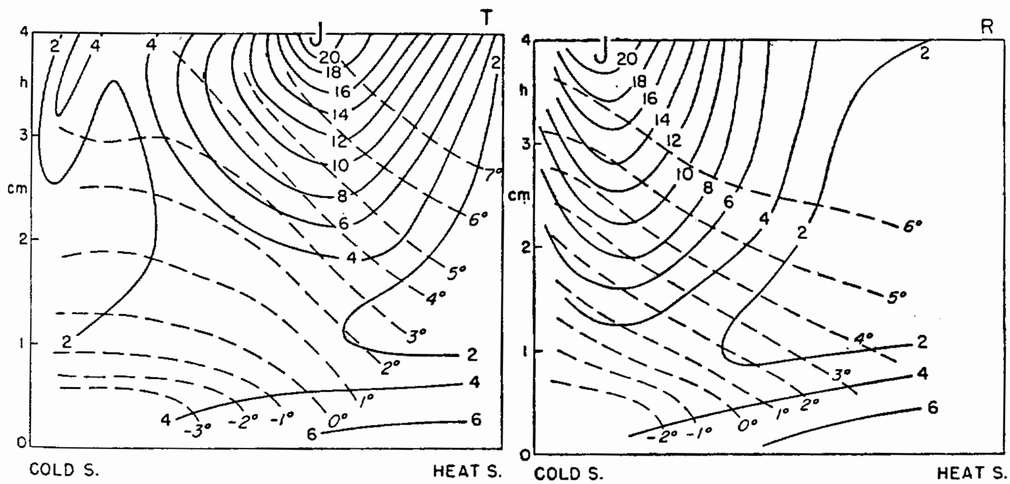
cylinder. The whole bottom thus becomes a momentum source. It is significant that in spite of this difference in arrangement *a replica of the atmospheric sea-level chart occurs,* and that altogether we see in Fig. 7 *the principal features of the atmospheric long and short waves merged into one simple basic pattern.* Clearly, momentum balance considerations should be very cautiously used in deriving insight into the reasons for the occurrence of the observed general circulation, since essentially the *same* circulation becomes established with such a *different* distribution of momentum sources and sinks.

7. Vertical cross-sections

Because of the great stability of the water the significant features of the temperature field are best brought out through subtraction of the mean temperature-depth curve of Fig. 5 (b) from the temperature cross-sections (Fig. 8 (a), (b)); cf. isotherms on Figs. 9 (b), (c). Along both trough and ridge the stability decreases from polar to equatorial



(a)



(b) (c)

Figure 9. (a) Vertical cross-section of speed of flow (per cent) at south-west inflection point. For location of section see Fig. 10. (b) Vertical cross-section of flow speed (per cent) and temperature (departure from 16°C) in trough (T). (c) Same as Fig. 9 (b) for ridge (R).

latitudes. In the trough, almost the whole meridional temperature gradient above the bottom layer is concentrated near the jet stream and, as is typical for the atmosphere, the jet-stream centre is located in the middle of the gradient near the zero anomaly line. Such a position of the core is also observed near the surface in the ridge. At greater depths in the ridge, however, the anomaly lines slope equatorward indicating greater vertical stability and smaller vertical slope of the isotherms than in the trough under the jet.

The wind-field at the south-west inflection point (Fig. 9 (a)) is spectacular and differs from atmospheric vertical sections only in the strong easterly velocities near the bottom. The horizontal cyclonic shear is much greater than the anticyclonic shear through a deep layer. Figs. 9 (b), (c) show vertical cross-sections of wind and temperature in trough and ridge.

8. Vertical motion

Quantitative observation of vertical displacements was not possible under the conditions of the experiment, but the main features of the calculated vertical motion pattern to be described were later verified qualitatively by dropping dye in the interior of the pan and following the motion of the coloured water visually. Vertical motions were computed with the assumption of conservation of heat, i.e., temperature. This assumption is clearly inapplicable near all four boundaries; there, however, the condition applies that the vertical motion must vanish. In the interior, molecular diffusion alone can produce changes in heat content. As the lapse rate is almost linear (c.f., Fig. 5 (b)) such diffusion can be shown to be without importance.

Because of the steady state, the substantial derivative

$$\frac{d}{dt} = c_r \frac{\partial}{\partial s_r},$$

where c_r is the three-dimensional velocity relative to the waves and s_r is along the three-dimensional relative streamline, (also a trajectory since the motion is steady in the wave coordinate system). c_r may be approximated by c_{rg}, the relative geostrophic velocity in the horizontal plane since we may expect the vertical velocities to be at least one order of magnitude less than the horizontal velocities. The relative streamline distance s_r however cannot be so treated. The ratio of vertical to horizontal extent of the water is 1 : 4, while in the atmosphere the corresponding ratio is 1 : 1000. Thus three-dimensional relative streamlines had to be constructed on constant-temperature surfaces for the dishpan. This was no easy task and several techniques of construction were tried out. Most satisfactory was Fultz's adaptation for the dishpan* of Montgomery's (1937) isentropic geostrophic potential plus further modification for relative motion. Some of the relative streamline maps computed by this method are shown in Fig. 16.

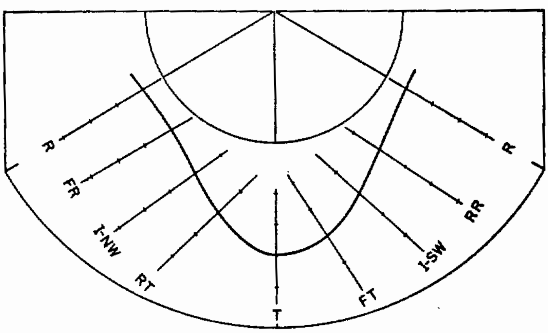


Figure. 10. Scheme of cross-sections normal to jet-stream axis on top surface.

* To be published.

The vertical velocity

$$w \equiv \frac{dz}{dt} = c_{rg} \frac{\partial z_T}{\partial s_r},$$

and after division by C_e

$$w' = c_{rg}' \frac{\partial z_T'}{\partial s'_r}, \qquad . \quad . \quad . \quad . \qquad (3)$$

where z_T is the height of the constant temperature surface and $s_r' = s_r/r_0$. Given the fields of c_{rg}', s_r' and the topography of the isothermal surfaces, w' can be computed. Such computations were made on seven surfaces 1°C apart. Some adjustment for vertical consistency proved necessary; all adjustments however were relatively minor and nowhere altered the computed pattern substantially. The results will be shown at first with vertical cross-sections (Fig. 11) as laid out in Fig. 10. The unit of vertical motion is per mille (of $r_0\,\omega$). Because the relative vertical extent of the dishpan is 250 times that of the atmosphere, all values must be divided by 250 for comparison. Thus 5 per mille is equivalent to 1 cm/sec, and we see from Fig. 11-13 that the magnitude of vertical speeds in the

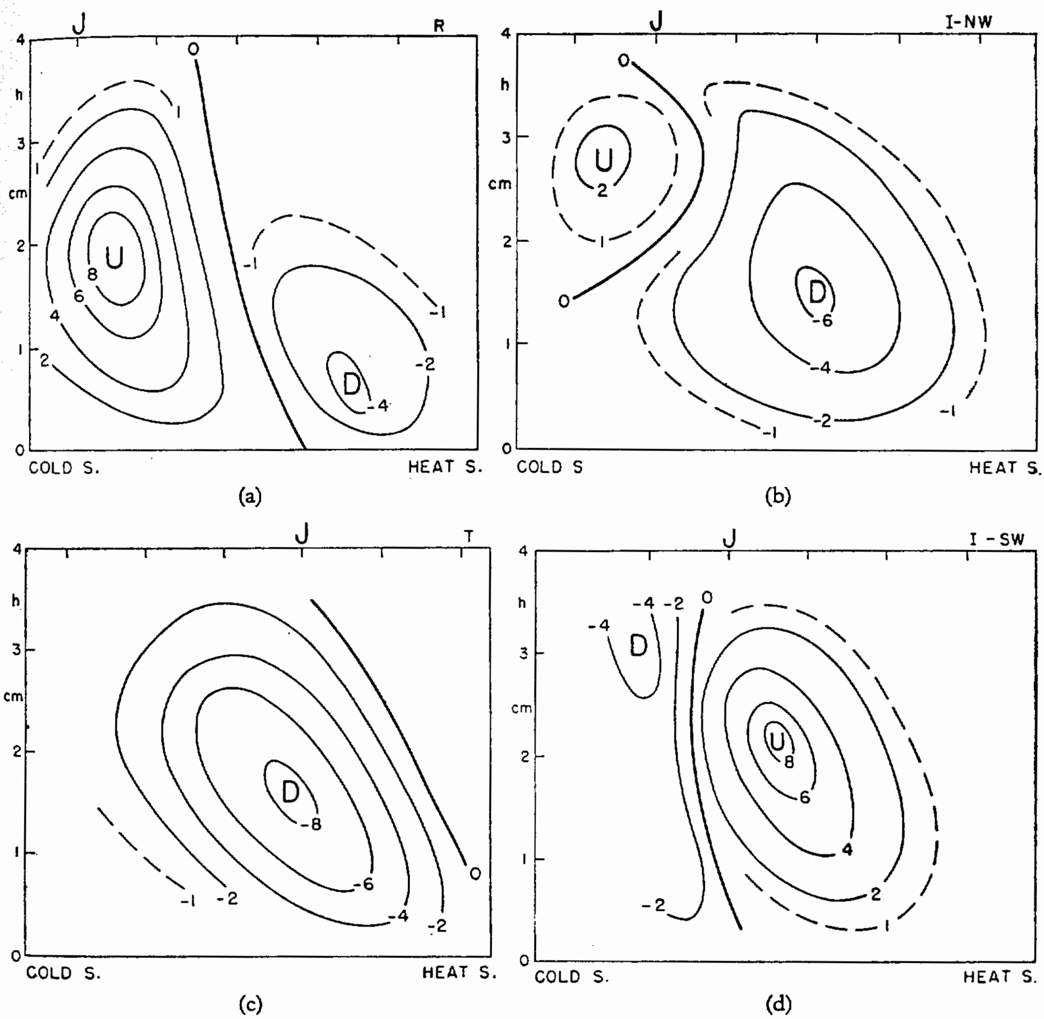


Figure 11. (a), (b), (c) and (d) Vertical cross-sections of vertical motion (per mille). For location of sections see Fig. 10.

dishpan, just as that of horizontal speeds, compares satisfactorily with the atmosphere. This method of comparison amounts to treating the vertical exaggeration by using total depth as the reference quantity for vertical distances and $z\omega$ instead of $r_0\,\omega$ for the vertical component of velocity. Rossby (1926) and Fultz *et al.* (1956) have discussed conditions necessary for this to be valid; mainly, that the hydrostatic approximation for the vertical equation of motion applies.

In the ridge (Fig. 11a) we observe ascending motion under the jet and descent in the warmer water toward the equatorial rim. Thus the circulation is solenoidally indirect over most of the cross-section; but the centre of ascent lies to the right of the jet-stream axis, looking downstream, so that the vertical circulation is direct at the core. The analysis has been extrapolated to top and bottom where the vertical velocity must vanish; it has been terminated a few millimeters from heat and cold sources where visual observation of coloured water shows strong ascent, or descent, in accord with expectations. This procedure has been followed on all cross-sections and charts.

The vertical circulation is definitely indirect about the jet axis in the north-westerly flow (Fig. 11 (b)). General descent takes place in the trough (Fig. 11 (c)), with the centre of descending motion situated under and slightly to the left of the jet stream core. Finally, at the south-west inflection point (Fig. 11 (d)) strong direct circulation prevails. In summary the sense of the vertical circulation about the jet-stream axis is not uniform. Direct circulation occurs in the south-westerly part of the flow and indirect circulation in the north-westerly part. However, the direct portion is the stronger and also occupies a longer distance along the axis. It is of interest that direct and indirect branches appear even though there is no concentrated velocity maximum along the core, hence no entrance and exit zones. These branches must therefore be related to the wave motion.

Fig. 12 shows a vertical cross-section along the core, obtained by averaging the vertical motion in each normal cross-section over a distance of 0·1 r_0 on both sides of the core. Centres of ascent and descent lie slightly east of the ridge and west of the trough. The zero lines extend nearly vertically at the north-west inflection point and somewhat west of the south-west inflection point. Broadly, descent takes place in the region of cyclonic flow curvature at the top surface, ascent in the region of anticyclonic curvature. In view of the flow pattern on the 1-cm surface (Fig. 7) one can also say that descent occupies the area of northerly flow at that level between cyclone and anticyclone, ascent the area of southerly flow, in excellent agreement with the atmosphere.

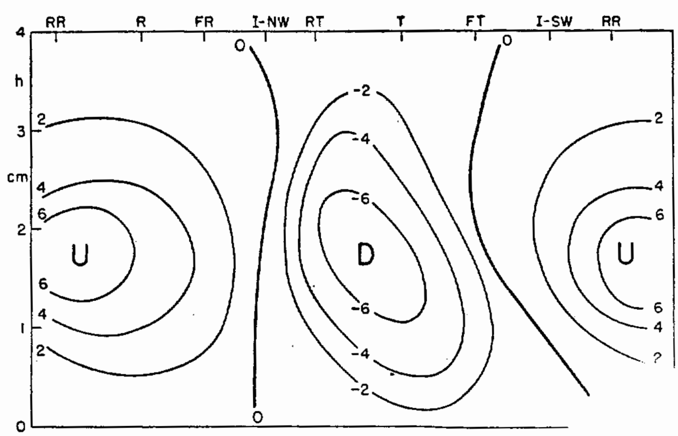


Figure 12. Vertical cross-section of vertical motion (per mille) following jet-stream core on top surface. Horizontal axis determined from Fig. 10.

In water, the level of non-divergence corresponds almost to the level of strongest vertical motion; according to Fig. 12 it is situated therefore in the mean slightly below the 2-cm surface which divides the mass in equal halves. This corresponds well with the average position of the level of non-divergence near 600-500 mb in the atmosphere. As in the atmosphere, however, non-divergence is not found near the centre of the mass everywhere, but, as shown by Fig. 11, there are sloping surfaces of non-divergence which reach from bottom to top of the pan. The slope is always poleward and, as suggested by the patterns of Fig. 13, the surfaces of non-divergence start near the equatorial rim of the pan and from there spiral toward the poleward rim with counterclockwise curvature.

Figs. 13 (a), (b), (c) and (d) depict the vertical motion on four horizontal charts; the jet-stream axis has been added for reference. We observe the features already noted from the vertical sections : strong ascent to the right and strong descent to the left of the axis in the south-westerly flow, and a weaker reverse circulation west of the trough. Vertical motion gradients are strongest at the position of the cyclone at 1 cm, and the areas of ascent and descent spiral inward cyclonically from the outer rim.

Since the temperature field is steady in the wave-coordinate system, horizontal advection must compensate for vertical temperature advection. Except near top and bottom the temperature increases upward everywhere, so that colder water is moved upward and warmer water downward by the vertical displacements. Steady state demands that relative to the waves horizontal advection of warm water takes place in the areas of ascent and advection of cold water in the areas of descent. Further, since the speed of the flow increases upward everywhere above the bottom layer of easterlies, the water accelerates as it rises and decelerates as it sinks. As in the atmosphere, individual particles do not conserve their energy; increases, or decreases, of potential and kinetic energy are coupled. It also follows that acceleration takes place in areas of relative horizontal advection of warm water and deceleration in areas of relative cold advection.

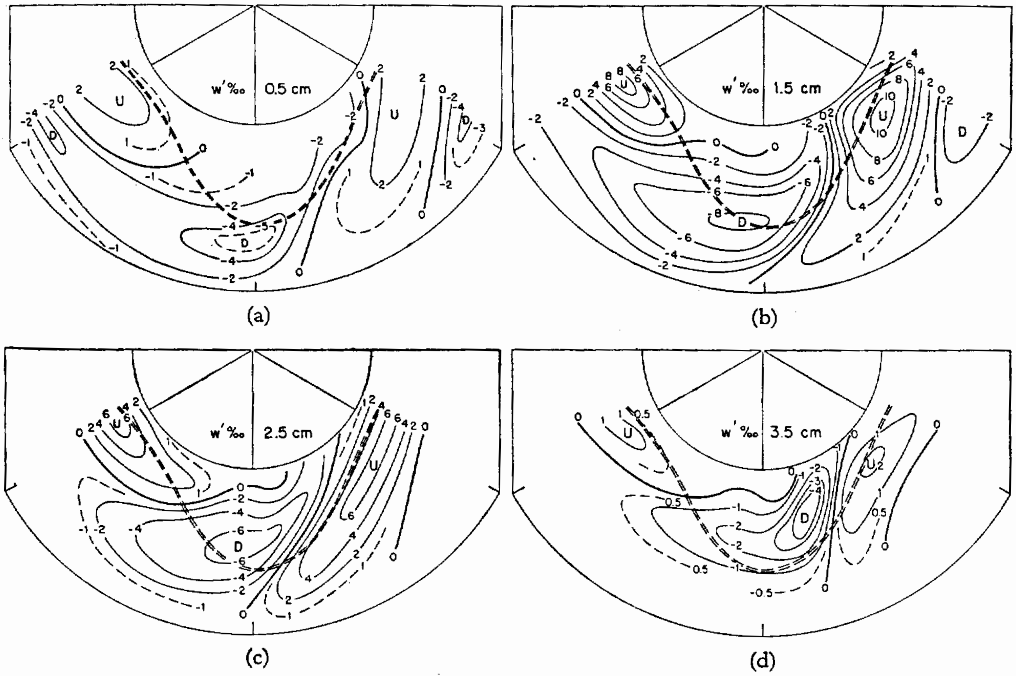
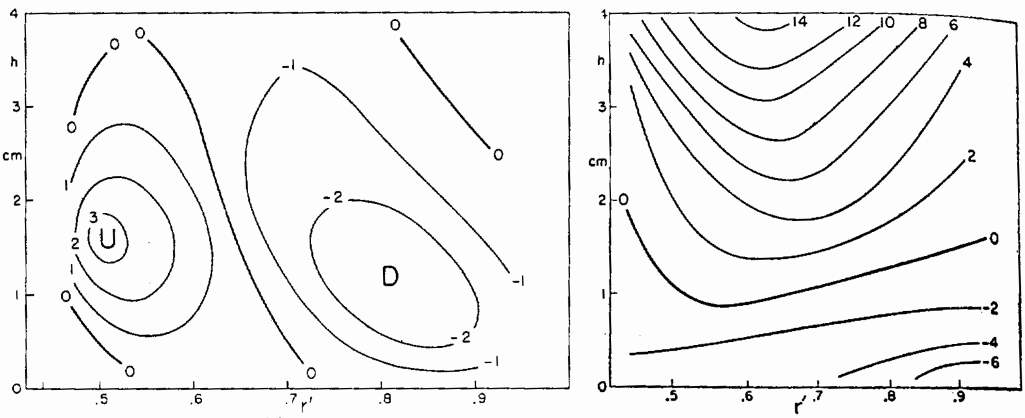


Figure 13. (a), (b), (c) and (d): Vertical motion (per mille) on four selected levels with heights indicated on diagrams. U denotes centre of ascent and D centre of descent; double-dashed line is jet-stream axis.

Regional distributions of temperature, wind and vertical motion are often summarized in general circulation studies in cross-section form after averaging around the globe. Fig. 13 shows that much caution is necessary in performing the averaging so that proper deductions can be drawn. A preferred method is zonal averaging. In Fig. 13 the area of strongest ascent lies poleward of the area of strongest descent. If one averages zonally, one will inevitably arrive at a picture with ascent poleward of the descent (Fig. 14 (a)). If the horizontal flow is also averaged zonally, the mean zonal flow distribution of Fig. 14 (b) is obtained. This leads to the apparent result that ascent takes place north of the strongest westerlies, and descent to their south; if Fig. 5 (a) is also considered one might reason that the cold water rises and the warm water sinks. When downward motion at the cold source and upward motion at the heat source are added to the pattern of Fig. 14 (a), the classical atmospheric three-cell structure emerges.



(a) (b)

Figure 14. (a) Zonally-averaged distribution of vertical motion (per mille) with radius and depth. (b) Zonally-averaged distribution of horizontal flow speed (per cent) with radius and depth. Distance r' at bottom as fraction of r_0.

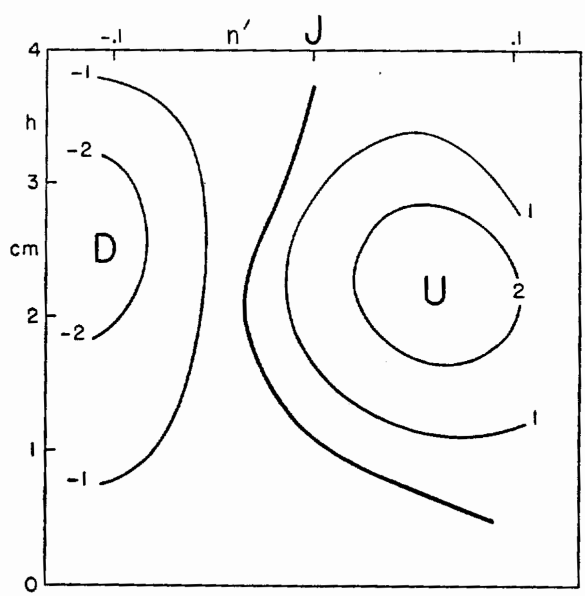


Figure 15. Distribution of vertical motion (per mille) averaged following jet-stream axis on top surface. Distance normal to jet axis expressed as fraction of r_0 at top (n').

Fig. 14 (a), of course, is not incorrect and truly represents the zonally-averaged distribution of vertical motion. As the foregoing has demonstrated, however, a simple interpretation by comparison with Figs. 5 (a) and 14 (b) is not permissible. The technique of zonal averaging has arisen basically from the concept of small perturbations superimposed on a broad zonal current. Both in dishpan and atmosphere, however, the perturbations are large and calculations based on coordinate systems implying small perturbations for their significance, lose much of their meaning. Analogous to Fig. 13, the ' reverse ' middle-latitude circulation cell of the atmosphere can be taken to signify little except the well-known fact that in cyclones the ascent above warm fronts takes place north of the descent behind cold fronts. As in the dishpan it does not follow that cyclones are energy-consuming systems; rather the reverse is known to be true. What does follow is that zonal averaging can, to great advantage for solving many problems, be replaced by averaging following the large perturbations, in our case the jet stream, as it traces out the long-wave pattern. When this is done, we obtain the vertical motion pattern of Fig. 15 which immediately suggests that the jet-stream zone may be a region of potential-energy release. This subject will be explored fully in a further report.

9. Vorticity field

The vorticity field was calculated to determine whether the theorem of conservation of potential vorticity is the dynamic law guiding the motion. We also hoped that this calculation would shed some light on an apparent difference between dishpan and atmosphere. Long waves occur in both systems. In the atmosphere their appearance has been ascribed to variation of the Coriolis parameter with latitude (Rossby 1940): in the dishpan however the Coriolis parameter is constant.

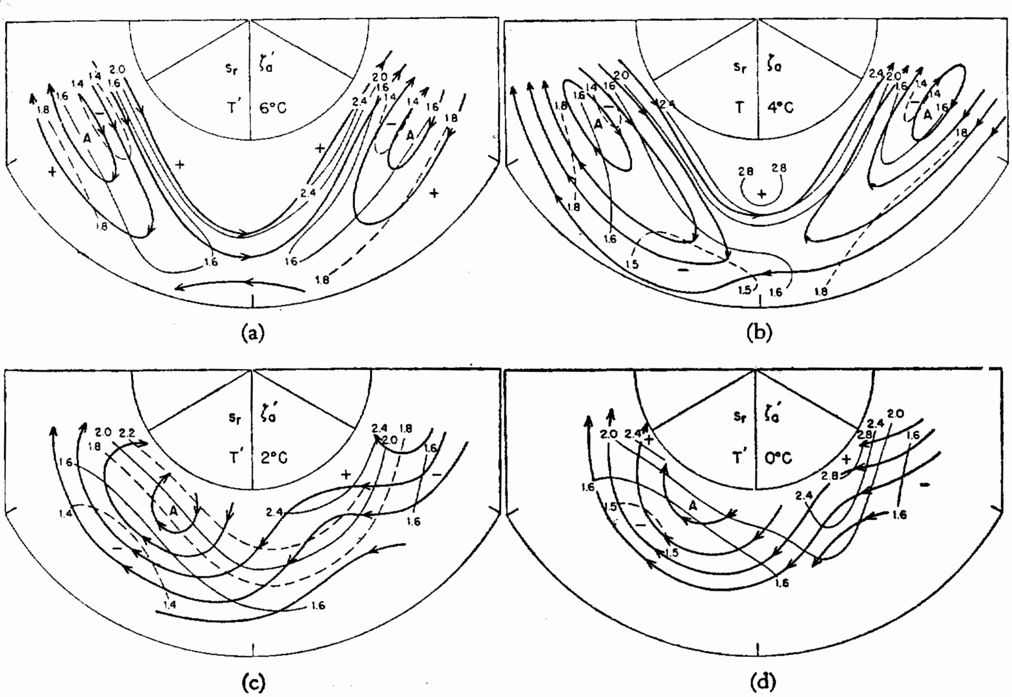


Figure 16. (a), (b), (c) and (d): Relative streamlines s_r (with arrows) and isolines of absolute vorticity ζ_a' (non-dimensional) on four selected constant-temperature surfaces. The value 2·0 denotes Coriolis parameter.

As various recent studies have shown, the vorticity in the atmosphere is logically computed on isentropic surfaces when the motion is adiabatic. Except for the top surface (Fig. 4), this is even more true for the dishpan because of the relatively great depth of the water. Figs. 16 (a), (b), (c), (d) shows analyses of absolute vorticity on four constant-temperature surfaces. Of these, the upper two (6° and 4°C) indicate conditions in the layer with waves, the lower two (2° and 0°C) in the layer with vortices. The relative streamlines have been added so that the motion of the water through the vorticity field can be followed. Since the relative streamlines are geostrophically computed, they are not quite comparable to Fig. 2 where the total motion, including non-geostrophic flow, is depicted. Hence the vorticity variation along the relative streamlines, while given approximately, will not be exact; this comment also holds for the preceding calculation of vertical motion.

The analysis could not be carried above the 3·5-cm level because of non-adiabatic motion near the top, so the 6°C and 4°C surfaces do not reach the jet-stream core. They enter the jet stream sufficiently to permit several deductions, however. Near the jet-stream axis the water conserves its absolute vorticity in the north-west flow; in the trough-line and to its east the vorticity increases somewhat, in accord with Fig. 4. The total vorticity variation is very small, however, actually within range of error of the analysis. It follows that with some approximation *conservation of absolute vorticity is the applicable dynamic law in the layer with wave motion*. This is surprising since at first sight one would judge from Fig. 1 that the vorticity of individual particles increases from ridge to trough, and decreases from trough to ridge. But, as seen from Fig. 8, the vertical slope of the isotherms is smaller in ridge than in trough. Therefore the anticyclonic shear equatorward of the jet-stream axis is smallest in the ridge because the constant-temperature surfaces pass through a smaller fraction of the vertical wind shear than in the trough (Figs. 9 (b), (c)). The increase of anticyclonic shear from ridge to trough compensates for the change from clockwise to counter-clockwise flow curvature. On the left of the jet-stream axis a similar consideration applies. There the constant-temperature surfaces cross the isotachs at a larger angle in ridge than in trough so that the cyclonic shear is greatest in the ridge, whereas the shear may be zero or even small anticyclonic in the trough (Fig. 9 (b)). It follows that variations of the slope of the isotherms along the jet-stream core in the dishpan take the place of variations of the Coriolis parameter in the atmosphere in order to maintain the same type of motion.

Equatorward of the jet-stream region marked vorticity changes occur. On the 6°C surface we find anticyclonic relative streamlines curving toward decreasing vorticity in the southwesterly flow. This is the region where strong ascent takes place as the water enters the jet stream from the equatorial side. West of the trough-line the descending water acquires higher vorticity. On the 4°C surface the same type of pattern is still present but no longer with the same clarity, as this surface drops well below the 2-cm level in the equatorial portions. A completely different regime prevails on the two lower surfaces. All relative motion is easterly. The current acquires cyclonic vorticity as it moves toward the area of the cyclone found on the 1-cm surface (Fig. 7). It loses vorticity during passage from cyclonic to anticyclonic centre.

These results are satisfactory and show the mechanism of maintenance of the vortices. Looking at upper and lower surfaces jointly, the water moves toward higher vorticity east of the cyclone at low levels and toward lower vorticity at high levels. The vertical motion is upward, with largest speeds at intermediate heights. West of the cyclone, inverse conditions obtain. Qualitatively, then, the *theorem of conservation of potential vorticity is applicable*. We also attempted to calculate the potential vorticity (ξ_p) directly from the relation $\xi_p = \xi_a/\Delta$, where Δ is the depth of the column considered. The potential vorticity should be constant along the relative streamlines. This calculation failed because Δ could not be measured with sufficient accuracy. The mean thickness of a column between two constant-temperature surfaces 1°C apart is only 0·4 to 0·5 cm (Fig. 5 (b)); the analysis of the topography of the surfaces can be considered correct only within 0·2 cm. This is a small error compared to the vertical extent of the temperature surfaces which is 2 to 3 cm in the layer with adiabatic motion. The error, however, has the percentual magnitude of the vorticity variations, precluding valid computations of potential vorticity.

Conclusion

This report has shown that the three-dimensional structure of long waves and jet stream in the dishpan, indeed the structure of all major features, resembles the atmosphere in basic pattern and relative magnitude of the motions to a high degree. In a further report (in course of preparation), the mechanisms of the general circulation, including production and maintenance of kinetic energy, will be discussed.

Acknowledgments

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